

Groundwater quality evaluation using GIS and water quality index in and around inactive mines, Southwestern parts of Cuddapah basin, Andhra Pradesh, South India

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ABSTRACT

The main purpose of this study is to evaluate the groundwater quality for human consumption by developing a water quality index for the southwest (SW) part of the Cuddapah Basin. In order to study this various physico-chemical parameters were analyzed, distribution maps were prepared using Geographical Information System (GIS). Except for fluoride and nitrate, most of the chemical parameters are within the allowable range, which is due to geogenic and anthropogenic sources, respectively. Most of the heavy metals were below the permissible limit except cadmium, which might be due to anthropogenic activities. The piper diagram shows that most samples belong to the Ca^{2+} - Mg^{2+} - HCO_3^- category. Rock dominance is the predominant hydrogeochemical factor controlling groundwater chemistry. According to WQI classification, 12% of samples are not potable due to anthropogenic activities. This study proved that the combined application of GIS and WQI is very effective in accessing groundwater quality.

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1. Introduction

Mining is one of the most important sectors to generate an economy for developing countries like India. The mining industry, such as gold, coal, barite, yellow stone, asbestos, limestone, etc., is developing rapidly and has become an incentive for the country's economic growth (Brindha et al., 2016). Active mines are generating inactive or abandoned mines, and polluting groundwater media without taking proper closer remedial actions. If a mine was inactive, the dewatering facility of the mine was ceased. In the monsoon season during heavy rainfall surface water may have entered through openings such as adits, fractures, and fissures of the underground mine; such mine water contains a significant concentration of heavy metals and total dissolved solids. This may have led to higher temperature and altered pH depending on the nature of ore body and the neutral local geochemical conditions belongs to alkaline mine water. It may also contain high pollutants higher than water quality standards and have a significant impact on human fitness and the environment. Groundwater quality problems may also be due to contamination of natural and anthropogenic activities are added to pollutants into the groundwater system (Muralidhara Reddy et al., 2019; Adimalla, 2019; Adimalla et al., 2019; USEPA, 1993). The groundwater quality increasingly corresponding to depth and also alters owing to the composition of the dissolved rocks,

physical, chemical, biological, seepage and rock sediment activities. Adverse changes in groundwater quality have occurred in recent years owing to an increased in urbanization, industrialization and agricultural activities across the globe. Owing to overuse, freshwater has become a scarce asset and water is being polluted leading to adverse health implications (Elango et al., 2012; Qian and Li, 2011; Qian et al., 2012; Li and Qian, 2018, Li et al., 2018a, 2018b; Wu et al., 2014, 2017; Zhang et al., 2018; Elango et al., 2003).

Agriculture is a significant economic sector in India, 60% of the people are dependent on it. In India more than 20 million private & government tube wells are active and is utilized for drinking, irrigation purposes (Datta and Tyagi, 1996). In India 80% of groundwater is utilized for domestic purposes while 45% of groundwater is used for irrigation. Agriculture contributes for 45% gross of the national product by majority of the people (Kumar et al., 2005; Sunitha and Sudharshan Reddy, 2019; Singh, 1983). Groundwater quality information provides us the past geological history of the rocks, changes in groundwater level, groundwater recharge and discharge movement, and aquifer storage capacity of the aquifer.

The water quality index (WQI) is defined by the combined influence of different parameters of water quality as a reflection and ranking. Water quality indexes are the most valuable ways to align common public and policy maker's knowledge on water quality to control water quality (Satyajit et al., 2020; Chimankpam et al., 2019; Prasad et al., 2019; Suvarna et al., 2018). The majority of the researchers use these indices for drinking water suitability (Adimalla and Venkatayogi,

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2018). However, in the Cuddapah basin, Andhra Pradesh state a few research activities has been noted especially on groundwater quality and their quality indices. Sreedhar and Nagaraju (2017) reported that weathering of rock-forming minerals strongly reflects the water chemistry in Thummalla palli area, Kadapa district. Sunitha and Sudharshan Reddy (2019) conducted an investigation on groundwater chemistry of the Lakkireddypalli and Ramapuram region in Kadapa district reported that quality of groundwater mostly depends on mechanisms of silicate weathering and partially carbonate dissolution. The above research workers reported that the mechanism of the water chemistry mainly depends on the weathering of rock-forming mineral dissolution. Therefore, it is very important for the planning and implementation of water management strategies to assess the groundwater quantity & quality and provide a database. Therefore, assessing the quantity and quality of groundwater and providing a database are very important for the planning and development of water management technologies. In view of this, the main purpose of the analysis is to highlight groundwater quality in order to establish drinking water quality indicators in the southwestern part of the Cuddapah Basin using GIS techniques.

2. Study area

Cuddapah basin is of crescent - shaped with a maximum width of 145 m (in the middle) and 440 km long covering an area of 44,000 sq. km. This region falls in SW part of Cuddapah basin (latitude 14° 21' 00" N; longitude 78° 22' 30" E) occupies an area of 408.6 Sq.km and

falls in survey of India toposheet No.57J/3, J/7 (Fig. 1). Southwestern part of the Cuddapah basin rest on archaen gneissic complex covering greenstone belt of Gadwal, Kadiri, Velligallu with a predominant unconformity. The study area belongs to the most significant Purana basins in Southern India. Stratigraphically study area consists of Papagni and Chitravati groups of Lower Cuddapah Supergroup (Nagaraja Rao et al., 1987). Lithologically, this region consists of conglomerate, shale, basalt, dolomite, massive limestone, and some of the basic intrusions (Sills) (Fig. 3). The major geomorphic features in this part are denudational hills, pediment, pediplain and structural hills. Groundwater found in weathered zone of Papaghni and Chitravathi rock formations. The water present in dug wells in this part is chiefly due to the presence of numerous fractures and joints as well as fissures zones in this rock types. Massive limestone, shale and quartzite can serve as good aquifers, and the permanent groundwater level is usually shallow. Water level depth varies from 10 to 50 m (CGWB, 2013). Paddy and sweet lime are important crops; banana plantations also grow seasonal fruits (sweet lime, lemon) (Sudharshan Reddy et al., 2020). The annual average rainfall is 600–650 mm, and the influence of southwest wind and northeast monsoon is also noted.

2.1. Inactive/abandoned mines

Study area exposed to numerous open pit and underground inactive mines (barite, yellow ocher, asbestos) and it makes nuisance to local habitats. Inactive mines possess serious threat to health and safety of

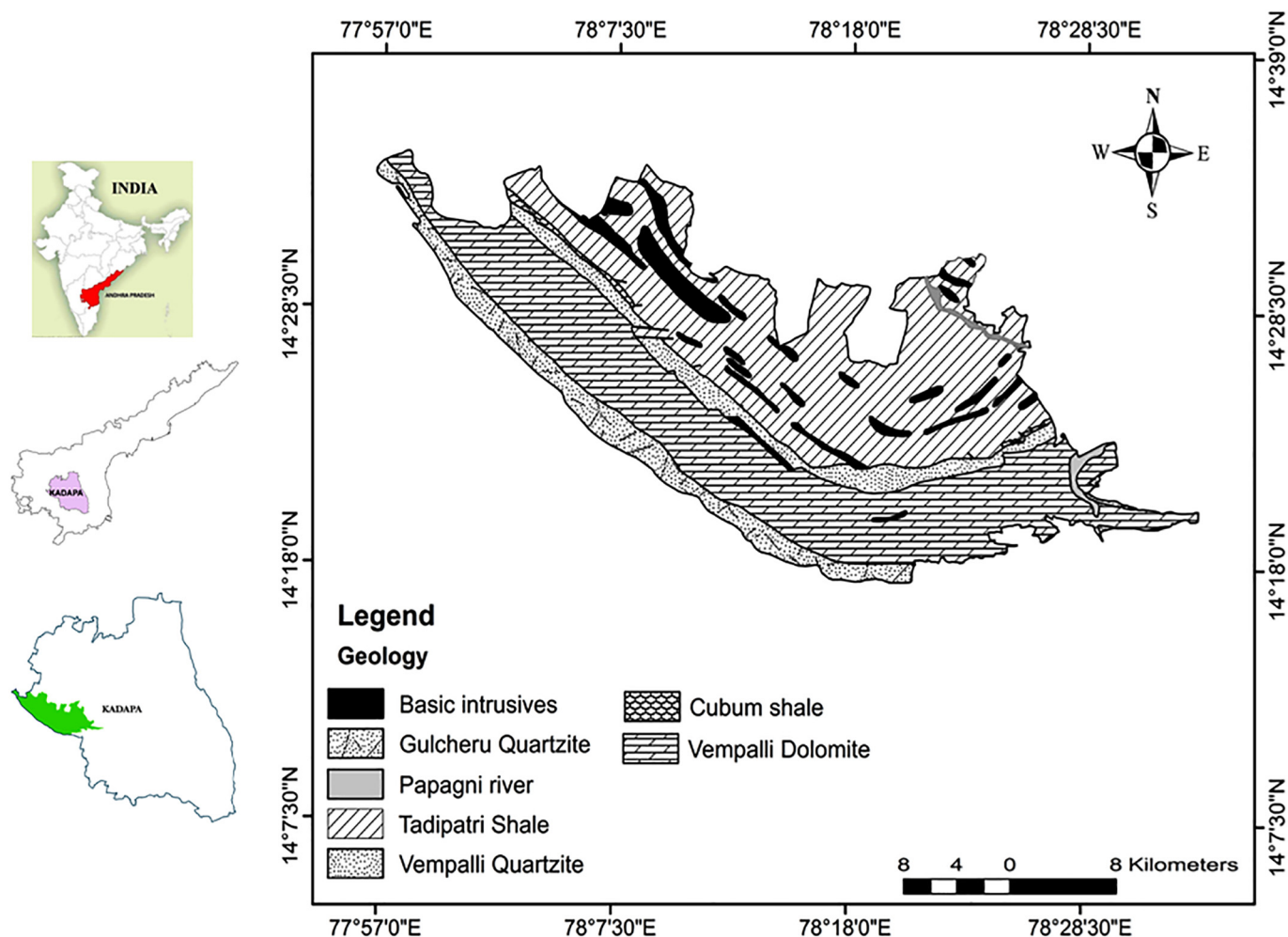


Fig. 1. Location map of the study area.

the surrounding human communities especially without undertaking the proper cessation procedures and necessary safety and greenery measure. Inactive related impoundments, waste piles were disturbed in the environment of clogging the streams and contaminated by surface and groundwater media. Most of the inactive/abandoned land-mines are located near residential areas along with agricultural land in the study area. Because of this excessive domestic garbage and agricultural garbage (chemical fertilizers, pesticides, etc.) are dumped into inactive mining areas. Due to the heavy rainfall in the rainy season after the monsoon, agricultural wastewater will flow into inactive mine impoundment and flow to groundwater through infiltration and return for irrigation and drinking purposes. Inactive mine related dangers are unprotected high walls, unsealed vertical openings, posing a threat to the health and safety of surrounding human communities (Sudharshan Reddy et al., 2020).

3. Materials & methods

3.1. Sampling & chemical analysis

Fifty Groundwater samples were collected from bore wells in pre-cleaned and well-dried polyethylene bottles in September 2018 (Fig. 2). Then water was filtered with 0.45 Millipore membrane. Physicochemical parameters were measured with various instruments like pH is measured with pH meter, electrical conductivity, and total dissolved solids (Conductivity cell CD-10) were measured with water analyzer 371 model field kit. Samples were stored at 4 °C. Major cations like Ca^{2+} , Mg^{2+} , CO_3^{2-} , HCO_3^- , Cl^- were determined by titration methods, Na^+ , K^+ were determined using the flame photometer (Systronics

Model No-128); anions like SO_4^{2-} , NO_3^- were analyzed by spectrophotometric technique and F^- was analyzed by using ion-selective electrode (Orion 4 star ion meter, Model: pH/ISE) (Hem, 1991; Raghunath, 1987; APHA, 2012) (Table 1).

In order to measure the analytical accuracy between total cations like Na^+ , K^+ , Ca^{2+} , Mg^{2+} and total anions like CO_3^{2-} , HCO_3^- , Cl^- , SO_4^{2-} , NO_3^- and F^- denoted in milliequivalent per liter (meq/L) for each sample, ionic balance error (IBE) was checked by the following equation:

$$\text{IBE} = \frac{\sum \text{Cations} - \sum \text{Anions}}{\sum \text{Cations} + \sum \text{Anions}} \times 100$$

The ionic balance error result (IBE) was within the acceptable limit of $\pm 3\%$ (Domenico and Schwartz, 1990).

3.2. Geographical information system (GIS) analysis

GIS is crucial for understanding water quality analysis with production of maps. The present study was carried out with an aid of topographical sheets and Arc GIS 10.3. Study area map prepared by 1:50,000 scale and applying the WGS 1984 coordinate system to prepare base maps. Global position system (GPS) readings were used for demarcating the sample collection mapping in the study area. Extended module of Arc GIS 10.2 was applied to prepare various distributions maps on hardness, pH, and other major cations and anions by spatial interpolation technique through inverse distance weight (IDW) (Adimalla and Taloor, 2020).

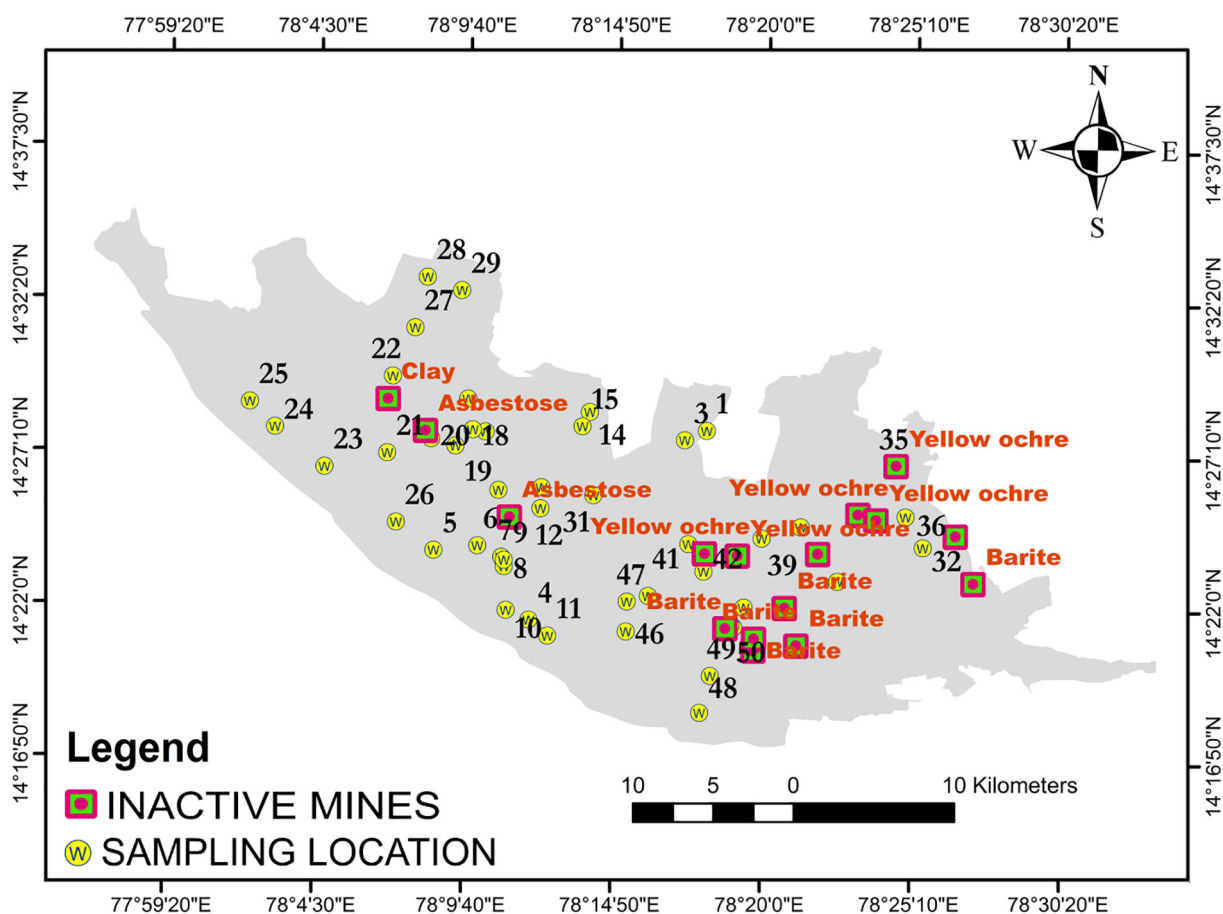


Fig. 2. Sample collection map.

Table 1
Range of chemical parameters and their comparison with the WHO and the Indian standards for drinking water quality.

Chemical parameter	Concentrations ions		Ranges of standards		Sample numbers exceeding permissible limit (%)	
	Range	Mean	(desirable - permissible)		WHO (2011)	BIS (2012)
			WHO (2011)	BIS (2012)		
pH	7.4–8.9	8.1	7–9.2	6.5–8.5	Nil	Nil
EC (µS/cm)	115–2920	1260	–	–	Nil	Nil
TDS (mg/L)	110–1987	676	500–1500	500–2000	2	Nil
TH (mg/L)	75–530	261	100–500	300–600	4	Nil
Na ⁺ (mg/L)	11–429	85	50–200	–	4	Nil
K ⁺ (mg/L)	0.5–37	6	–	–	Nil	Nil
Ca ²⁺ (mg/L)	6.9–103	46.9	75–200	75–200	Nil	Nil
Mg ²⁺ (mg/L)	11.5–152	52	30–150	30–100	2	6
CO ₃ ²⁻ (mg/L)	0.001–60	23.7	–	–	Nil	Nil
HCO ₃ ⁻ (mg/L)	24–366	188.00	300–600	300–600	Nil	Nil
SO ₄ ²⁻ (mg/L)	10.6–224	63.00	200–600	200–400	Nil	Nil
Cl ⁻ (mg/L)	0.03–322	80.00	250–600	250–1000	Nil	Nil
NO ₃ ⁻ (mg/L)	0.14–589	78.0000	45	45–100	42	14
Br (mg/L)	0.001–0.745	0.28	–	–	Nil	Nil
F ⁻ (mg/L)	0.45–4.7	1.5000	0.5–1.5	1.0–1.5	36	36

3.3. Water quality index estimation

Generally, WQI has a widely adopted equation for communicating and understanding the overall quality of the groundwater (Mouna et al., 2012; Pradhan et al., 2001). The term WQI is applied to find out groundwater suitability for drinking purposes (Subba Rao, 1997). The relative weight method was used to calculate groundwater quality in this region (Brown et al., 1970).

WQI calculations constitute three consecutive steps (Yidana and Yidana, 2010). “Weight assigning” is of first step, in which a weight (wi) has been assigned to each of the 14 parameters based on its relative importance to the overall drinking water quality. The second step includes “calculation of relative weight” by the following equation:

$$Wi = \frac{wi}{\sum_{i=1}^n wi} \quad (1)$$

“Rating of quality (qi)” includes the third step, as determined by the following equation:

$$qi = \left(\frac{Ci}{Si} \right) \times 100 \quad (2)$$

where the concentration of each parameter is represented as Ci in individual water sample, and Si is the specified value of a individual parameter prescribed by WHO. Lastly, the Wi and qi were used to

determine the Sli for each parameters and therefore WQI can be determined by the equation as shown below:

$$Sli = Wi \times qi \quad (3)$$

$$WQI = \sum_{i=1}^n Sli \quad (4)$$

where Sli is the sub-index of each parameter.

4. Results & discussions

4.1. Assessment for groundwater quality for drinking purpose

Groundwater quality evaluation for drinking purposes and its suitability was determined by its applicability based on the specific standards proposed by World Health Organization WHO (2011). The distribution maps of physicochemical parameters are depicted in Fig. 1 (a–n).

pH
pH range of groundwater samples was 7.4 mg/L and 8.9 mg/L, with an average of 8.1 mg/L (Table 2). In the study area, the highest percentage of groundwater samples is alkaline. It has no direct impact on human health, but the increase and decrease of pH will affect the health effects on various organisms in the ecosystem (Umer et al., 2019). The pH change of groundwater in this area may be caused by weathering and biochemical reactions of plagioclase feldspar in sedimentary rocks. (Ravikumar et al., 2011).

Electrical conductivity (EC)

The highest level of EC in drinking water was approved as 1500 µS/cm (WHO, 2011). Generally, EC depends on the temperature, concentration and ions present in groundwater. When the EC concentration increases, the corresponding TDS concentration also increases. At 25 °C, the EC range of groundwater is 115 to 2920 µS/cm, with an average value of 1260 µS/cm (Table 2). Fig. 1b depicts that 20% of the samples exceeded the EC regulatory limit observed in the western part of the study area. It was found that the higher EC may be caused by human activities and high mineral content or large amounts of total dissolved salts in groundwater.

Total dissolved solids (TDS)

The concentration of TDS in groundwater varies in different geological structures and their mineral solubility (WHO, 1984). In this area, the TDS value varies between the minimum value of 110 mg/L and the maximum value of 1987 mg/L, with an average value of 676 mg/L (Table 2).

Table 2
Groundwater classification based on TH, TDS.

Parameter	Classification	Range	Number of samples	% of Samples	References
Total	Fresh water	<1000	40	80	US
Dissolved Solids (TDS)	Slightly saline	1000–3000	20	20	Geological Suvery (2000)
	Moderately saline	3000–10,000	–	–	
	High saline	10,000–35,000	–	–	
Total	Safe	<75	Nil	–	Sawyer et al. (2003)
Hardness (TH)	Moderate	75–150	9	18	
	Hard	150–300	25	50	
	Very Hard	>300	16	32	

Table 3
Correlation of water quality parameters.

	pH	EC $\mu\text{S}/\text{cm}$	TDS mg/L	TH mg/L	Ca mg/L	Mg mg/L	K mg/L	Na mg/L	Cl mg/L	Br mg/L	No3mg/L	So4mg/L	TA mg/L	F mg/L
pH	1													
EC $\mu\text{S}/\text{cm}$	−0.39	1												
TDS mg/L	−0.38	0.99*	1											
TH mg/L	−0.47	0.80*	0.80*	1										
Ca mg/L	−0.07	0.67*	0.67*	0.62*	1									
Mg mg/L	−0.51	0.75*	0.75*	0.98*	0.48	1								
K mg/L	−0.13	0.29	0.29	0.20	0.21	0.18	1							
Na mg/L	−0.22	0.72*	0.72*	0.44	0.46	0.39	0.47	1						
Cl mg/L	0.21	0.23	0.23	0.16	0.20	0.13	−0.12	0.13	1					
Br mg/L	0.06	0.07	0.07	0.09	0.05	0.08	−0.19	−0.005	0.74*	1				
No3mg/L	0.27	0.07	0.08	−0.00	0.02	−0.01	−0.25	0.03	0.48	0.47	1			
So4mg/L	−0.05	0.30	0.31	0.22	0.31	0.18	−0.13	0.22	0.49	0.46	0.33	1		
TA mg/L	−0.24	0.32	0.31	0.11	−0.07	0.15	−0.11	0.20	−0.30	−0.26	0.08	−0.06	1	
F mg/L	0.10	−0.18	−0.17	−0.20	−0.16	−0.18	−0.29	−0.21	−0.24	−0.11	0.09	−0.13	0.35	1

WHO (2011) recommended that the maximum allowable TDS is 500 mg/L and the maximum is 1500 mg/L, so most samples fall within the allowable limit. According to the recommendations of the United States Geological Survey (2000) (Table 3), 80% of the samples belong to the freshwater category, and 20% of the groundwater samples belong to the mild saltwater category. Changes in TDS may be caused by carbonate deposition, salt leaching from the soil, and agricultural runoff (Fig. 1c).

Total hardness (TH)

Total hardness is defined as the sum of temporary hardness and permanent hardness. Water with hardness higher than 300 mg/L is unsafe and may cause heart disease and kidney-related health problems (Mouna et al., 2012). According to the international standards of the World Health Organization, the maximum allowable limit for drinking TH is 500 mg/L, and the ideal limit is 100 mg/L. The total hardness varies from 75 mg/L to 530 mg/L (Table 2). The acceptable limit of total hardness (TH) is 300 mg/L to 600 mg/L (WHO, 2011). According to Sawyer et al. (2003) groundwater is considered as safe: <75, moderate to hard: 75–150; Hard 150–300; Very hard: >300 and 80% of groundwater is rated as hard to very hard category and is noted in western & eastern part of the study region (Table 3) (Fig. 1d); it may be due to leaching of calcium and magnesium bicarbonate ions through recharge.

Major cations and anions

For cations and anions, the dominant modes of ions are $\text{K}^+ > \text{Mg}^{2+} > \text{Ca}^{2+} > \text{Na}^+$ and $\text{HCO}_3^- > \text{Cl}^- > \text{NO}_3^- > \text{SO}_4^{2-} > \text{CO}_3^{2-} > \text{F}^- > \text{Br}^-$, respectively. Carbonate (CO_3^{2-}) and bicarbonate (HCO_3^-) in groundwater are between 0.01 mg/L to 60 mg/L and 24 mg/L to 366 mg/L, with average values of 23 mg/L and 188 mg/L, respectively (Table 1). The HCO_3^- content in groundwater is relatively high; it has no direct adverse health effects on humans. Most of the groundwater samples fall within the desirable limit of 300 mg/L (WHO, 2011). The permissible limit of bicarbonate in drinking water is 150 mg/L (WHO, 1984). Very limited samples exceed the prescribed limit of bicarbonate due to carbonated rich sedimentary formations (dolomites) are contributes to carbonate and bicarbonate to the groundwater in the study region.

The chloride concentration in this area varies from 0.03 mg/L to 322 mg/L, with an average value of 80 mg/L (Table 1). The ideal limit for chloride in drinking water is 250 mg/L and the allowable limit is 1000 mg/L (World Health Organization, 2011). The higher chloride concentration in the southeastern part (322 mg/L) may be due to the wasted leakage of yellow stone (Fig. 1g), or it may be due to the leaching of the upper soil by industrial and household activities and dry climates (Adimalla and Venkatayogi, 2018; Sunitha and Sudharshan Reddy, 2019). All groundwater samples were below the prescribed limit. Calcium (Ca^{2+}) and magnesium (Mg^{2+}) contribute to the hardness of water. The content of calcium varies from 6.9 mg/L to 103 mg/L, and the value of magnesium varies from 11.5 mg/L to 152 mg/L (Table 1). The maximum allowable limit of calcium is 200 mg/L (WHO, 1983).

The high Ca^{2+} and Mg^{2+} concentrations found in the middle of the area may be due to the dissolution of carbonate rocks and alkaline intrusive rocks, which may be attributed to the cations exchange between iron and magnesium minerals in the rocks of the study area. Sodium (Na^+) and potassium (K^+) ions generally found in rocks and soils contained sodium compounds and are easily dissolved in groundwater; it is not harmful at normal levels. However when exceeding the permissible limits may harm to human health like hypertension, heart disease, or kidney problems. Sodium varied from 11 mg/L to 429 mg/L and the values of potassium ranged from 0.5 mg/L to 37 mg/L (Table 1). Potassium ion is mostly held by clay particles in soils. High sodium and potassium in groundwater in this region might be due to irrigation practices, leaching of potassium through soil profile into groundwater, and sewage effluent. Sulfates are found generally as barite (BaSO_4), epsomite ($\text{Mg SO}_4 \cdot 7\text{H}_2\text{O}$), and gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) minerals which can easily dissolved in groundwater (Greenwood and Earnshaw, 1985). The maximum contaminant level of sulfur in groundwater is 600 mg/L. High sulfates cause laxative effects on humans and impart unpleasant taste to drinking water. All of the groundwater samples were below the permissible limit of sulfate in this region.

Nitrate concentration in this region varies from 0.14 mg/L to 589 mg/L with a mean of 78 mg/L (Table 1). The main causes of nitrates (NO_3^-) in groundwater are the unrestricted use of nitrogen fertilizers, animal manure and the inability to use established water and soil management methods, septic tanks, and improper disposal of domestic waste (Canter, 1996). However, the high nitrate concentration in this area may also be due to the pollution of idle/abandoned mines. These impoundments acted as strategic reservoirs of agriculture waste; and directly in contact with groundwater table and are easily contaminated by NO_3^- ions into groundwater resources. Higher NO_3^- concentrations in drinking water can cause gastric cancer in adults, which is blue baby syndrome in infants (Bao et al., 2017). Bromide concentration in the study area varies from 0.0014 mg/L to 0.745 mg/L with an average of 0.28 mg/L (Table 1). Naturally bromide concentration low in the nature, except shales (up to 24 mg/kg). Bromide (Br^-) is found in tracer levels in the form of negative monovalent elements in all groundwater media because it is a halogen element (Carpenter, 1978; Rittenhouse, 1961; Edmunds, 1996). Chloride and bromide are inorganic compounds highly soluble in groundwater. Focus on bromide levels in groundwater is because of two main reasons (1) resolve the history of water; (2) used an artificial tracer for groundwater studies (Davis et al., 2003).

Fluoride (F^-) concentration in this region ranges between 0.4 mg/L to 4.7 mg/L with a mean of 1.5 mg/L (Table 1). 30% of the samples exceeding the prescribed limit of fluoride (1.5 mg/L) (WHO, 2011). Higher fluoride concentration might be derived from basic flows along with dolerites and carbonate minerals in this region (Brindha and Elango,

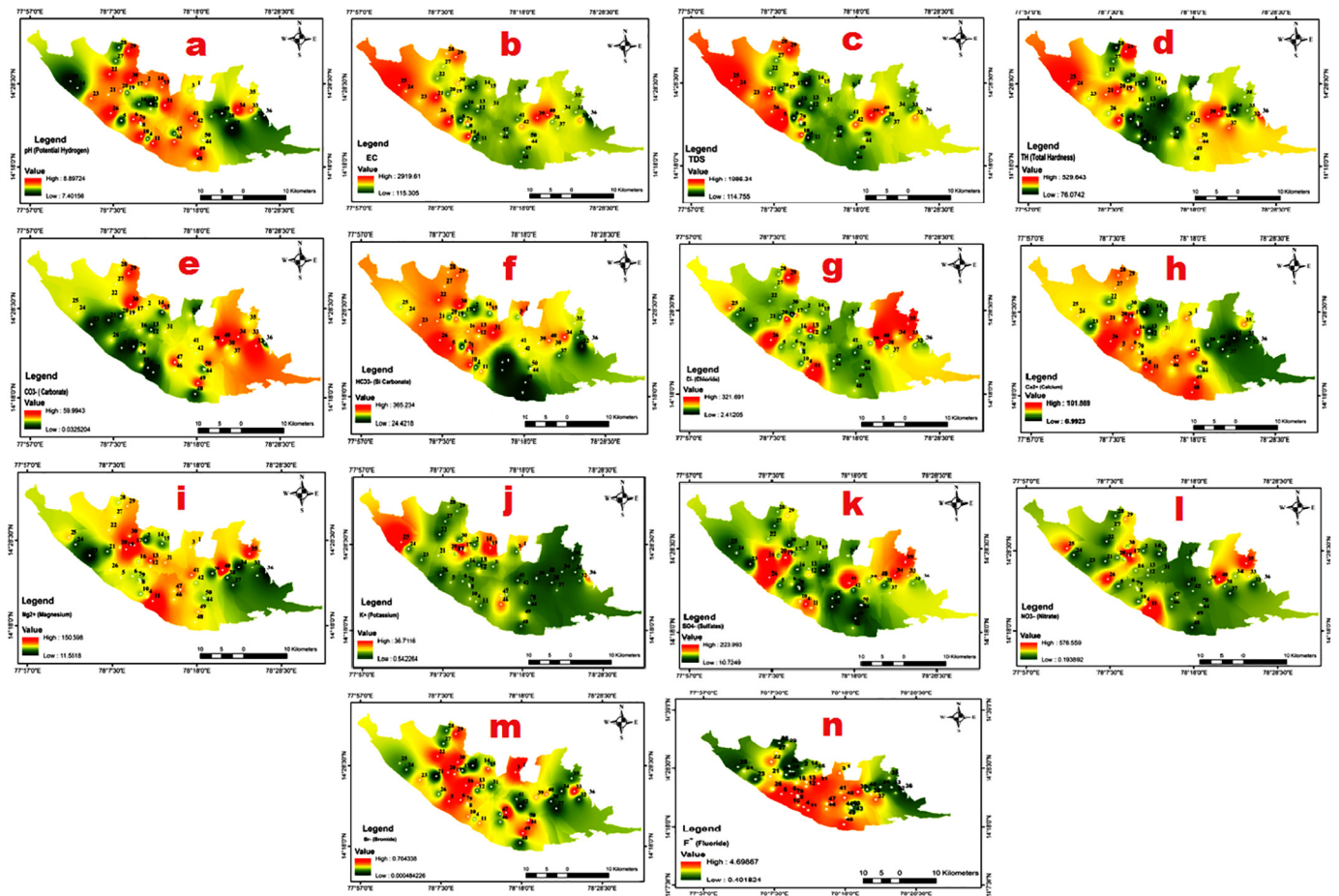


Fig. 3. [a–n] spatial distribution maps of individual physical parameters.

2013; Sudharshan Reddy et al., 2020). Geogenic sources like apatite, clay, biotite and also longer contact with the aquifer media under alkaline environment is also one of the key factors to cause higher concentration fluoride in this area (Subba Rao et al., 2002; Gizaw, 1996; Frengstad et al., 2001; Carrillo-Rivera et al., 2002; Jawed Akhtar and Rashid Umar, 2019; Raju et al., 2019). The residents of these villages rely only on groundwater for drinking, while the people in the study area are exposed to high levels of fluoride pollution in the groundwater. Fluoride as an isomorphous ion its replaced with the OH^- position (Brindha and Elango, 2011) due to a similar ionic radius of both fluoride and hydroxyl ion (F^- 133 pm, OH^- 140 pm) (O'Donnell et al., 1973). The groundwater of this region is alkaline and observed that an increase in

alkalinity made a similar increase in the fluoride concentration (Rao, 2009). In the alkaline environment F^- ions can be easily liberated and OH^- and F^- ions have similar radius and easily exchange with each other. The presence of fluoride bearing minerals and their interaction with water in a dry climate is considered as the major cause of fluoride concentration. It can be concluded that fluoride bearing water is usually high in alkalinity and low in hardness, chloride and sulfate (Muralidhara Reddy et al., 2019).

4.1.1. Spatial distribution of hydrochemical parameters

The spatial maps depict that majority of the hydrogeochemical parameters of this region are within the permissible limits, where as

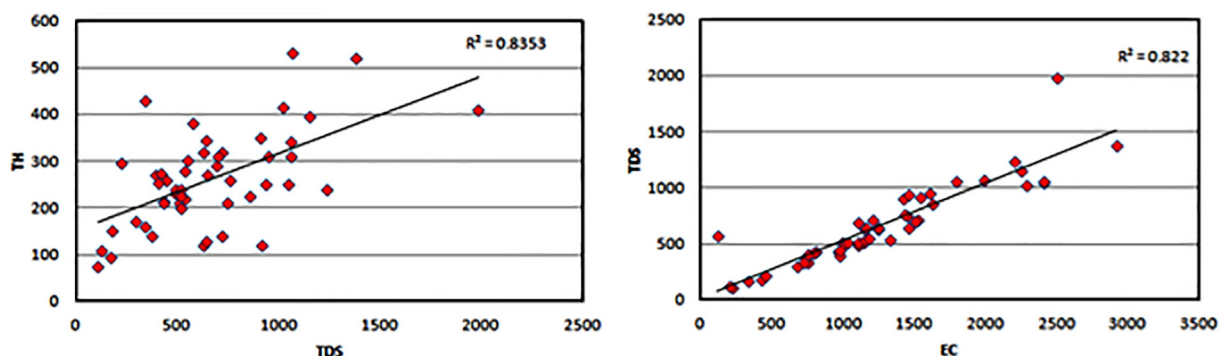


Fig. 4. Correlation matrix between (i) TH Vs TDS (ii) TDS Vs EC.

northwest and middle portion of this region has a higher concentration (Fig. 3 a–n). Lithologically north portion of this region is dominated by gulcheru quartzite and dolomite, middle portion by quartzite, dolomite, and shale. Groundwater chemistry is significantly dependent on dissolution of these rock formations and ion exchange reactions in northwest, middle portion of this region. Anthropogenic activities very common in local communities in the study region might have contributed to chloride, sulfate, and nitrate contamination (Fig. 3g, k, m). Intensive irrigation activities that include huge application of agrochemicals on farms is very common in this local community and is very well observed in this area (Sudharshan Reddy et al., 2020; Sunitha and Sudharshan Reddy, 2019). High fluoride concentration (>1.5 mg/L) were noted in the middle portion of this region (Fig. 3n); perhaps due to basic flows (dolerite) in contact with carbonate minerals leading to ion exchange mechanism, water-rock interactions and mineral solubility.

4.1.2. Correlation of the physico-chemical parameters of the groundwater

A Pearson correlation coefficient was applied to find out the relationship among numerical variables. It exhibits a correlation analysis of the ionic concentration of water between different physico-chemical parameters. It can be applied for understanding the geochemical process of the groundwater and from Table 3, it can be illustrated that EC is showing positive correlation with TDS and TH (Fig. 4). A strong correlation between magnesium and total hardness used to find out the mechanisms of carbonate dissolution in semi-arid region (Subramani et al., 2005).

4.2. Hydro-geochemical facies

Hydrogeochemical regime of groundwater is represented by plotting major cations and anions in the piper trilinear diagram. Generally, the Piper diagram consists of 3 distinct fields including the left hand side had cation triangular, the right hand side had anion triangular and the top one is a diamond-shaped field. Geochemistry of the groundwater can be studied from this Piper's plot which is further categorized into four groups viz., $\text{Ca}^{2+}\text{-Mg}^{2+}\text{-SO}_4^{2-}$, $\text{Na}^+\text{-K}^+\text{-Cl}^-\text{-SO}_4^{2-}$, $\text{Na}^+\text{-K}^+\text{-Cl}^-\text{-HCO}_3^-$ and $\text{Ca}^{2+}\text{-Mg}^{2+}\text{-HCO}_3^-$. Piper diagram (Fig. 5) reveals that most of the samples belong to hydro chemical facies $\text{Ca}^{2+}\text{-Mg}^{2+}\text{-HCO}_3^-$ category in this region. The Piper diagram (Fig. 5) shows that groundwater consists majorly of Ca-HCO_3 (Calcium-Bicarbonate) and Ca-Mg-Cl (Calcium-Magnesium-Chloride) type. Among cations magnesium (Mg^{2+}) and sodium (Na^+) are dominant, and among anions bicarbonate (HCO_3^-) is major dominant. However, a major number of samples fall in Ca-HCO_3 and Ca-Mg-Cl type, which exhibits the carbonate rock dissolution and silicate rock weathering (Adimalla and Venkatayogi, 2018; Panagiotis et al., 2019; Raju et al., 2019).

4.3. Evolution of groundwater chemistry mechanism

Gibbs diagram is plotted in order to evaluate the sources of dissolved chemical components in the area, such as evaporation dominance (ED), rock dominance (RD), precipitation dominance (PD) in this region. In order to know the chemical reaction between the composition of

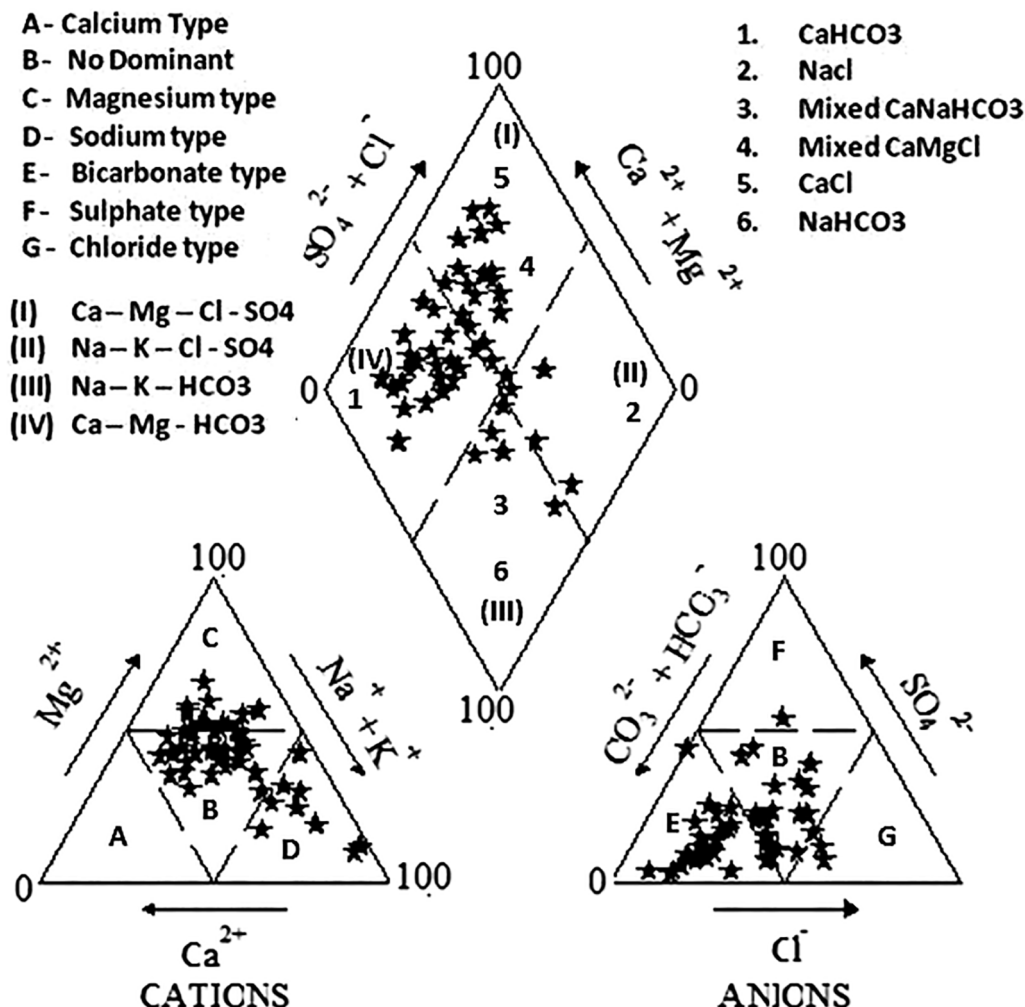


Fig. 5. Hydrochemical facies of groundwater of the SW part of Cuddapah basin, South India.

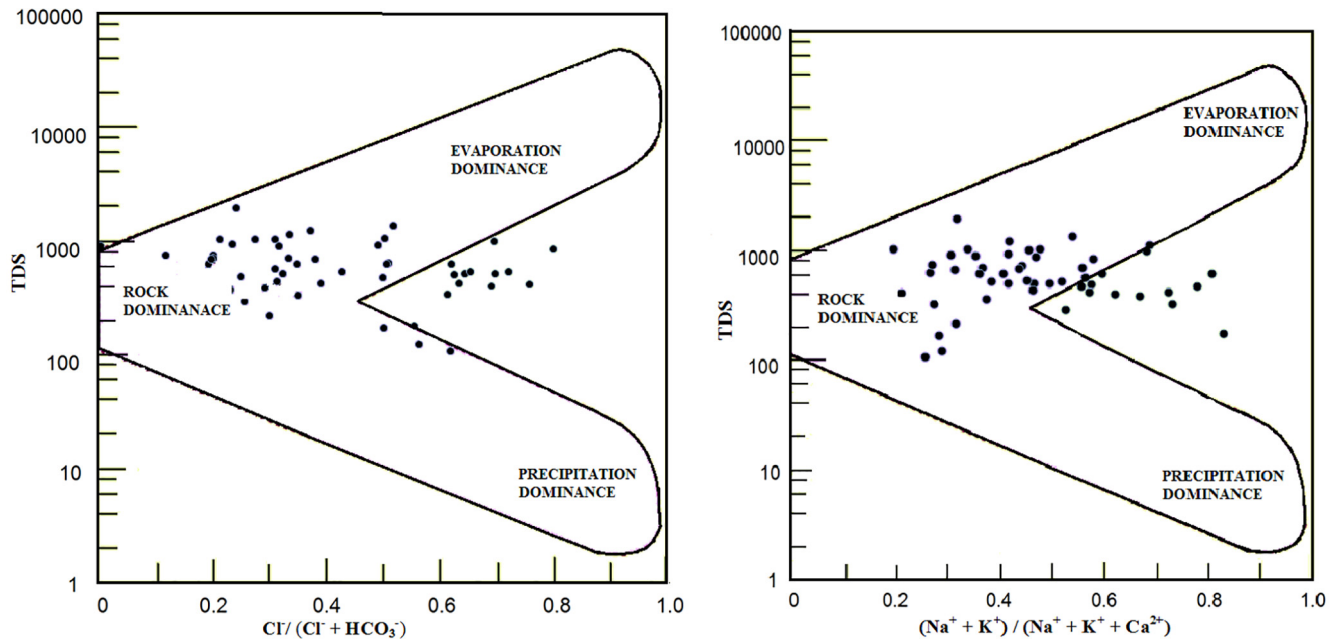


Fig. 6. Gibbs plot for controlling factors of groundwater quality showing rock dominance (rock-water interaction) in SW part of Cuddapah basin, South India.

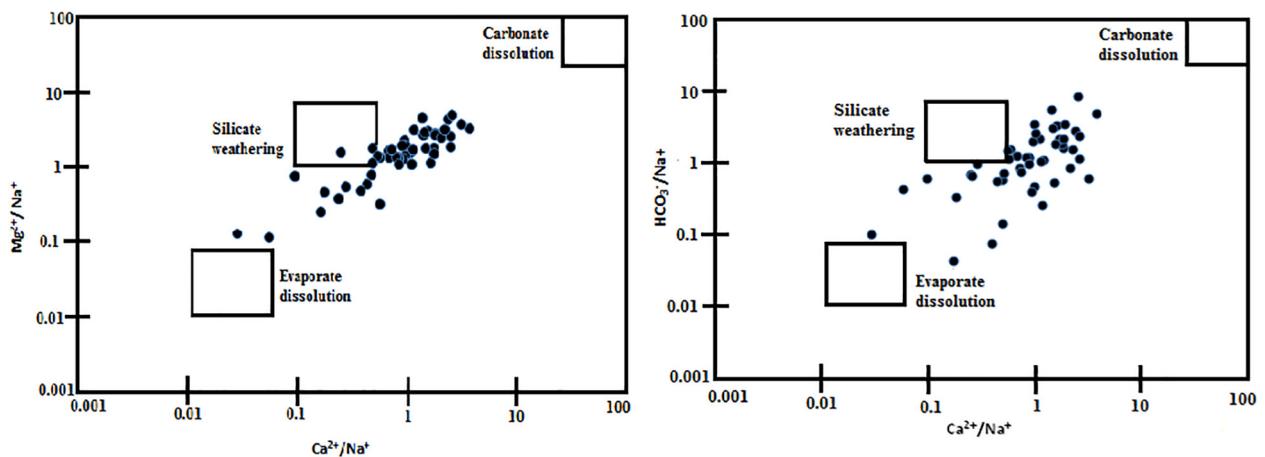


Fig. 7. Bivariate plots a. $\text{Ca}^{2+}/\text{Na}^+$ vs. $\text{Mg}^{2+}/\text{Na}^+$ b. $\text{Ca}^{2+}/\text{Na}^+$ vs. $\text{HCO}_3^-/\text{Na}^+$.

water and controlling mechanism of the natural water chemistry, Gibbs diagram is plotted using graphical plots of anions $(\text{Cl}^-)/(\text{Cl}^- + \text{HCO}_3^-)$ and cations $(\text{Na}^+ + \text{K}^+)/(\text{Na}^+ + \text{K}^+ + \text{Ca}^{2+})$ against TDS (Fig. 6). It reveals that most of the groundwater samples fall in rock dominance and some samples were partially inclined towards precipitation dominant; it suggested that groundwater chemistry is determined by rock water interactions and influenced by rock weathering and evaporation dominance which is further influenced by anthropogenic activities, leading to increase in chloride, sodium, and TDS in groundwater. According to this study, the rock–water interaction is one of the predominant factors determining groundwater chemistry in this region.

4.4. Normalized bivariate plots

This classification includes several lithological processes by weathering and dissolution processes controlling groundwater chemistry, and it is clear by plots of $\text{Ca}^{2+}/\text{Na}^+$ versus $\text{HCO}_3^-/\text{Na}^+$ and $\text{Ca}^{2+}/\text{Na}^+$ versus $\text{Mg}^{2+}/\text{Na}^+$ (Fig. 7). Groundwater chemistry is mostly

Table 4
Relative weight per parameter.

Chemical parameters	Weight(wi)	Relative weight(Wi)	Si	Ci	qi	Sli
pH	3	0.07	8	8.2	102.5	7.5
EC $\mu\text{S}/\text{cm}$	4	0.09	1500	724	48.2	4.7
TDS mg/L	2	0.04	600	375	62.5	3.04
TH mg/L	2	0.04	500	140	28	1.3
Ca mg/L	3	0.07	75	45.6	60.8	4.4
Mg mg/L	3	0.07	50	54.7	109.4	8.0
K mg/L	3	0.07	12	12.6	105	7.6
Na mg/L	3	0.07	200	33.4	16.7	1.2
Cl mg/L	3	0.07	200	50.6	25.3	1.8
NO_3 mg/L	5	0.12	10	28.3	283	34.5
SO_4 mg/L	5	0.12	250	25.6	10.24	1.2
F mg/L	5	0.12	1.5	1.09	72.6	8.8
	41	1				84.4

Table 5

Range of chemical parameters and their comparison with the WHO and the Indian standards for drinking water quality.

Chemical parameters	Ion concentrations	Highest permissible limit for water	% of samples exceeding permissible limit
	Range	World Health Organization (2006)	World Health Organization (2006)
Cd (µg/L)	4.7–20.5	5	12
Cr (µg/L)	1.9–4.2	50	Nil
Fe (µg/L)	15.8–43	300	Nil
Li (µg/L)	1.6–31.8	200	Nil
Mn (µg/L)	0.2–31.8	400	Nil
Ni (µg/L)	0–96	70	Nil
Sr (µg/L)	39.4–1122	25,000	Nil
V (µg/L)	3.9–30	50	Nil

Table 6

WQI at individual sampling stations.

S. No	Villages	WQI	Water quality status
1	Yerrapalli	84	Good
2	Achavalli	62.6	Good
3	Putrayunipeta	87	Good
4	Brahmanapalli	158.3	poor
5	Nallagonduvaripalli	132.9	poor
6	Nallapureddy Palli	119.4	poor
7	Chandragiri	194.6	poor
8	C.Kottapalli	125.6	poor
9	Mallikarjunapuram	78.8	Good
10	Motnuthanapalli	97.5	Good
11	Kanampalli	595.6	Water un suitable for drinking
12	Bramhanapalli	100.8	poor
13	Peddarangapuram	124.8	poor
14	R.Thummalapalli	106.9	poor
15	Lilapuram	60.58	Good
16	Ippatla	109.8	poor
17	Teranamapalli	101.1	poor
18	Akulavaripalli	815.4	Water un suitable for drinking
19	Chinnakudala	120.2	poor
20	Ramanuthalapalli	87.9	Good
21	Gunakanapalli	81.5	Good
22	Lingala	55	Good
23	Murrainuthala	66	Good
24	Degopalli	54.5	Good
25	Eguvapalli	281.5	very poor
26	Ambakapalli	301	Water un suitable for drinking
27	Bonala	116.2	poor
28	Ankevaripalli	516	Water un suitable for drinking
29	Kamasamdram	209	very poor
30	Peddakudala	103.9	poor
31	Pulivendula	120.6	poor
32	Pamuluru	59.1	Good
33	Ayyavaripalli	167.6	poor
34	Alavalapadu	127	poor
35	Nagur	367.8	Water un suitable for drinking
36	Ammayagaripalli	100.3	poor
37	Dhuggannagaripalli	57.5	Good
38	Chagaleru	77.4	Good
39	Gollalaguduru	119.9	poor
40	Pernapadu	410.9	Water un suitable for drinking
41	Pedajutur	62.1	Good
42	Kuppaguntlapalli	96.4	Good
43	Vemula	83.5	Good
44	Buchayyagaripalli	71.6	Good
45	Velupula	137.6	poor
46	Velupula	95.6	Good
47	Mobbuchintala Palli	86.4	Good
48	Rachakuntapalli	59.3	Good
49	Bommayyagaripalli	137.2	poor
50	Bochayyagaripalli	122.6	poor

influenced by silicate weathering and partially carbonate dissolution. In Southwestern part of Cuddapah basin predominant rock types are quartzite, shale with intrusions composing of feldspar (KAlSi_3O_8 – $\text{NaAlSi}_3\text{O}_8$ – $\text{CaAl}_2\text{Si}_2\text{O}_8$), quartz (SiO_2) are the abundant source of silicate minerals in this part. Calcium carbonate and silica are the most common cementing materials in such types of rocks. Therefore calcium, magnesium, sodium, and potassium might have significantly contributed through silicate weathering. Similar findings have been observed in silica-rich lithogenic areas (Brindha et al., 2016; Muralidhara Reddy et al., 2019).

5. Heavy metals in groundwater

Heavy metals are a significant group of pollutants which can create substantial harm to the environment when they are above their concentrations. It can be leached in to groundwater; it can bond with semi-permanently soils and taken up by plants, which later affect on human health. Due to the lack of undertaking the proper cessation adequate management like necessary safety planning, and poor waste management of inactive mines; it makes major sources of environment and of the water pollution. The ionic dominance pattern of heavy metals is in the order of $\text{Sr} > \text{Ni} > \text{Li} > \text{V} > \text{Cd} > \text{Mn} > \text{Fe} > \text{Cr}$, respectively. The statistical analysis of heavy metals in the study region was shown in Table 5. The results of chemical analysis depict that heavy metals were below the detection limits except cadmium. Cadmium (Cd) concentration in this region ranges between 4.7 µg/L to 20.5 µg/L. 12% of the samples exceeding the permissible limit of 3 µg/L of cadmium (WHO, 2011) due to anthropogenic activities such as huge usage of phosphate fertilizers and pesticides in the agricultural land of the study region (Mostafa et al., 2017).

6. Water quality index (WQI) of the study area

The water quality index is the essential calculated method to evaluate the water quality trend information. It provides data on water quality in a single value. Mostly it is done from suitability for human consumption. WQI calculation is followed by three steps (Horton, 1965; Pradhan et al., 2001; Yidana and Yidana, 2010; Mouna et al., 2012). The first step is based on its significant effect, for 13 parameters (such as pH, TDS, TH, Ca^{2+} , Mg^{2+} , K^+ , Na^+ , Cl^- , Br^- , SO_4^{2-} , NO_3^- , F^-) each parameter is assigned a weight (wi) about primary health. According to the maximum weight, 5 is allocated to parameters such as nitrate, sulfate and fluoride, and the minimum weight 1 is allocated to total dissolved solids, total hardness. The second and third steps are the calculation of relative weight and grade quality respectively (Table 6). Computed WQI values are represented in (Table 4). WQI values vary from 54.5 to 815.4 with a mean of 165 (Table 5). Based on water quality index results this region has been classified as five classes as excellent, good, poor, very poor, unsuitable to drinking purpose. According to this classification, 42% of samples are good, 42% belong to poor, and 4% fall in very poor category. 12% of samples are unsuitable for drinking purposes in this region (Table 7). As per WQI classification, eastern and southern parts of this region were not suitable for drinking purposes (Fig. 8).

Table 7

Drinking water quality classification Add reference for the table (Muralidhara Reddy et al., 2019).

Class	WQI Value	Water Quality Status	% of samples
I	< 50	Excellent	Nil
II	51–100	Good	42
III	101–200	Poor Water	42
IV	201–300	Very Poor Water	4
V	> 300	Water Un Suitable For Drinking	12

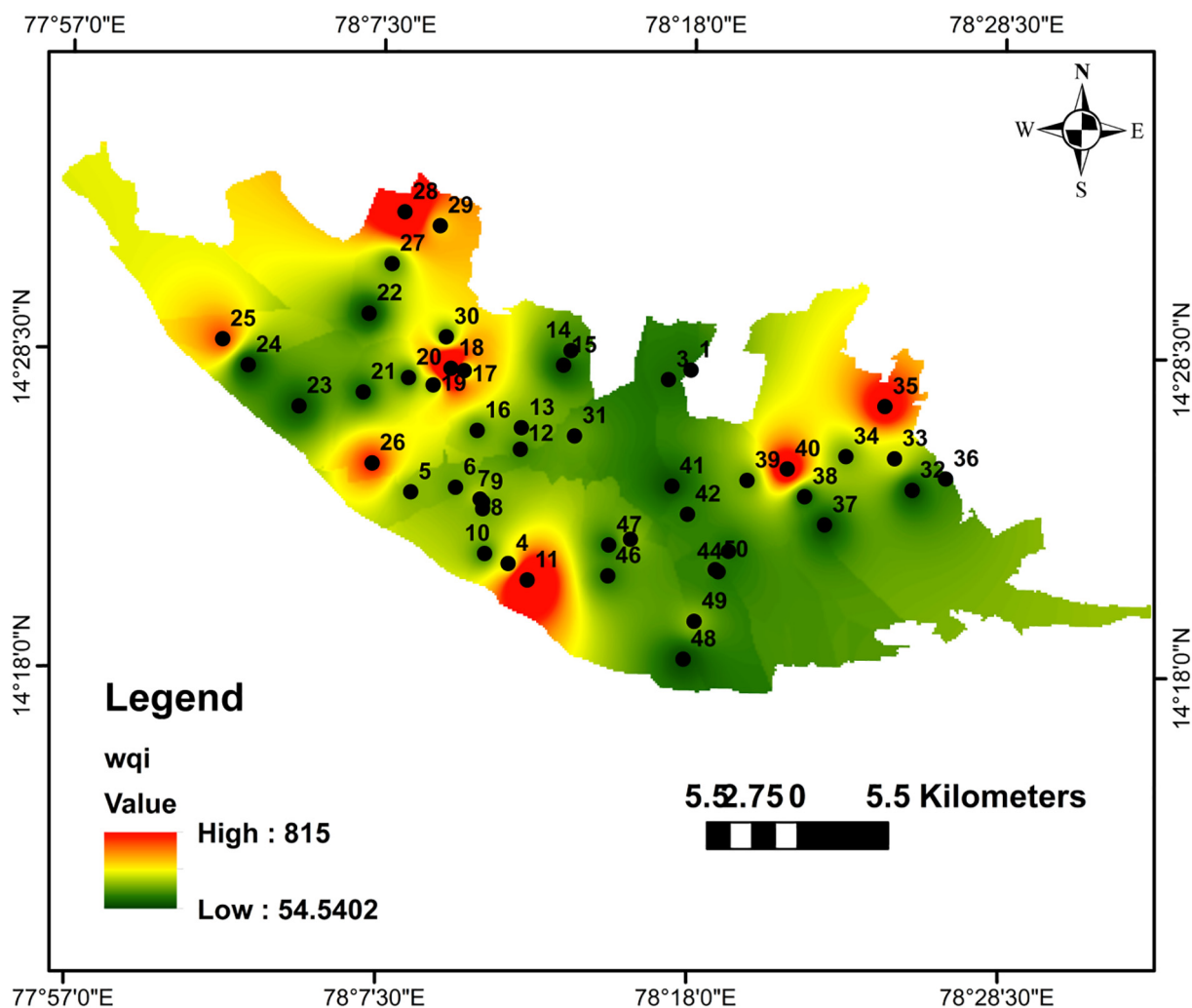


Fig. 8. Spatial distribution map of WQI.

7. Conclusion

The results of this study reveal that groundwater water is alkaline in nature. Most of the chemical parameters were within the permissible limit except fluoride and nitrate. 34% of groundwater samples are above the allowable limit for fluoride. Higher fluoride concentration might be derived from basic flows along with dolerites and carbonate minerals in this region. Geogenic sources like apatite, clay, biotite and also longer contact with the aquifer media under alkaline environment is also one of the key factors to cause higher concentration of fluoride in this area. 42% of the groundwater samples are above the allowable limit for nitrate. Unrestricted usage of nitrogen fertilizer, animal manure, and unavailability of established soil and water management methods, septic tanks, inappropriate removal of household garbage. Most of the heavy metals were below the desirable limit except cadmium, which might have been contributed by anthropogenic activities. The ionic dominance pattern is in the order of $K^+ > Mg^{2+} > Ca^{2+} > Na^+$ and $HCO_3^- > Cl^- > NO_3^- > SO_4^{2-} > CO_3^{2-} > F^- > Br^-$ for cations and anions, respectively. Piper diagram reveals that most of the samples belong to $Ca^{2+}-Mg^{2+}-HCO_3^-$ in this region. Higher concentrations of fluoride and nitrate in groundwater may have serious effects on human health. Rock dominance is the predominant hydrogeochemical factor controlling the water chemistry. According to this study, the rock–water interaction is one of the most important factors controlling groundwater chemistry in this area. According to WQI classification, 12% of samples

are not potable. The WQI distribution map reveals that the western and southern parts of the region not potable for drinking purpose due to anthropogenic activities. The results of this study have proved that the comprehensive use GIS and WQI is very helpful in accessing groundwater quality and with a clear view of geographic area of groundwater quality, policy makers can plan better for optimum utilization and management of groundwater resources.

Declaration of Competing Interest

None.

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References

- Adimalla, N., 2019. Controlling factors and mechanism of groundwater quality variation in semiarid region of South India: an approach of water quality index (WQI) and health risk assessment (HRA). *Environ. Geochem. Health*, 1–28 <https://doi.org/10.1007/s10653-019-00374-8>.
- Adimalla, N., Taloor, A.K., 2020. Hydrogeochemical investigation of groundwater quality in the hard rock terrain of South India using Geographic Information System (GIS)

- and groundwater quality index (GWQI) techniques. *Groundw. Sustain. Dev.* <https://doi.org/10.1016/j.gsd.2019.100288>.
- Adimalla, N., Venkatayogi, S., 2018. Geochemical characterization and evaluation of groundwater suitability for domestic and agricultural utility in semi-arid region of Basara, Telangana State, South India. *Appl Water Sci* 8, 44. <https://doi.org/10.1007/s13201-018-0682-1>.
- Adimalla, N., Qian, H., Li, P., 2019. Entropy water quality index and probabilistic health risk assessment from geochemistry of groundwaters in hard rock terrain of Nanganur County, South India. *Geochemistry*. <https://doi.org/10.1016/j.chemer.2019.125544>.
- APHA, 2012. *Standard Methods for the Examination of Water and Wastewater*. 22nd edn. .
- Bao, Z., Hu, Q., Qi, W., Tang, Y., Wang, W., Wan, P., Chao, J., Yang, X.J., 2017. Nitrate reduction in water by aluminium alloys particles. *J. Environ. Manag.* 196, 666–673.
- BIS, 2012. *Indian Standard, Drinking Standard Specification (Second revision)*. IS: 10500. Bureau of Indian standard, New delhi.
- Brindha, K., Elango, L., 2011. Fluoride in groundwater: Causes, implications and mitigation measures. In: Monroy, S.D. (Ed.), *Fluoride Properties, Applications and Environmental Management*, pp. 111–136.
- Brindha, K., Elango, L., 2013. Geochemistry of fluoride rich groundwater in a weathered granitic rock region, Southern India. *Water Qual Expo Health* 5, 127–138. <https://doi.org/10.1007/s12403-013-0096-0>.
- Brindha, K., Paul, P., Touleleor, S., Somphasith, D., Elango, L., 2016. Geochemical characteristics and groundwater quality in the vientiane plain. Laos. *Expo Health*. <https://doi.org/10.1007/s12403-016-0224-8>.
- Brown, R.M., McClelland, N.L., Deininger, R.A., Tozer, R.G., 1970. A water quality index: Do we dare? *Water Sewage Work* 117, 339–343.
- Canter, L.W., 1996. *Nitrates in Groundwater*. CRC Press ISBN 0873715691.
- Carpenter, A.B., 1978. Origin and chemical evolution of brines in sedimentary basins. *Oklahoma Geol. Survey Circular* 79.
- Carrillo-Rivera, J.J., Cardona, A., Edmunds, W.M., 2002. Use of abstraction regime and knowledge of hydrogeological conditions to control high-fluoride concentration in abstracted. *Groundwater: San Luis Potosi basin, Mexico*.
- Central Ground Water Board, 2013. *Ministry of Water Resources Government of India Ground Water Brochure YSR District (Kadapa), Andhra Pradesh*.
- Chimankpam, K.E., Obialo, S.O., Johnbosco, C.E., Chinanu, O.U., Daniel, A.A., 2019. Multi-criteria approach to water quality and health risk assessments in a rural agricultural province, Southeast Nigeria. *HydroResearch* 2, 40–48. <https://doi.org/10.1016/j.hydres.2019.11.005>.
- Datta, P.S., Tyagi, S.K., 1996. Major ion chemistry of groundwater in Delhi area: chemical weathering processes and groundwater flow regime. *J. Geol. Soc. India* 47, 179–188.
- Davis, S.N., Moysey, S., Cecil, L.D., Zreda, M., 2003. Chlorine-36 in groundwater of the United States: empirical data. *Hydrogeol. J.* 11 (2), 217–227.
- Domenico, P.A., Schwartz, F.W., 1990. *Physical and Chemical Hydrogeology*. Wiley, New York, p. 824.
- Edmunds, W.M., 1996. Bromine geochemistry of British groundwaters. *Mineral. Mag.* 60, 275–284.
- Elango, L., Kannan, R., Senthil Kumar, M., 2003. Major ion chemistry and identification of hydrogeochemical processes of groundwater in a part of Kancheepuram district, Tamil Nadu, India. *J. Environ. Geosci.* 10, 157–166.
- Elango, L., Brindha, K., Kalpana, L., Faby Sunny Nair, R.N., Murugan, R., 2012. Groundwater flow and radionuclide decay-chain transport modeling around a proposed uranium tailings pond in India. *Hydrogeol. J.* 20, 797–812. <https://doi.org/10.1007/s10040-012-0834-6>.
- Frengstad, B., Banks, D., Siewers, U., 2001. The chemistry of Norwegian ground waters: the dependence of element concentrations in crystalline bedrock groundwaters. *Sci. Total Environ.* 277, 101–117.
- Gizaw, B., 1996. The origin of high bicarbonate and fluoride concentrations in waters of the main Ethiopian Rift Valley. *J. Afr. Earth Sci.* 22, 391–402.
- Greenwood, N.N., Earnshaw, A., 1985. Health risk from acid rain: a Canadian perspective. *Environ. Health Perspect.* 63, 155–168.
- Hem, J.D., 1991. *Study and Interpretation of the Chemical Characteristics of Natural Water*: USGS Professional Paper Book 2254. Scientific Publishers, Jodhpur.
- Horton, R.K., 1965. An index number system for rating water quality. *J. Water Pollut. Control. Fed.* 37, 300–305.
- Jawed Akhtar, A., Rashid Umar, J., 2019. Evaluation of hydrogeochemical characteristics and groundwater quality in the quaternary aquifers of Unnao District, Uttar Pradesh, India. *HydroRes.* 1, 36–47. <https://doi.org/10.1016/j.hydres.2019.01.001>.
- Kumar, R., Singh, R.D., Sharma, K.D., 2005. Water resources of India. *Curr. Sci.* 89 (5), 794–811.
- Li, P., He, S., Yang, N., 2018b. Ground water quality assessment for domestic and agriculture purpose in Yan'an city, Northwest China: implication to sustainable groundwater quality management on the loess plateau. *Environ. Earth Sci.* 77, 775.
- Li, P., Qian, H., 2018. Water resources research to support a sustainable China. *Int. J. Water Res. Dev.* 34 (3), 327–336. <https://doi.org/10.1080/07900627.2018.1452723>.
- Li, P., He, X., Li, Y., 2018a. Occurrence and health implication of fluoride and ground water of loess aquifer in the Chinese loess plateau: a case study of Tongheun, Northwest China. *Exposure Health*. <https://doi.org/10.1007/s12403-018-0278-x>.
- Mostafa, M.G., Helal Uddin, S.M., Haque, A.B.M.H., 2017. Assessment of hydro-geochemistry and groundwater quality of Rajshahi City in Bangladesh. *Appl Water Sci* 7, 4663–4671. <https://doi.org/10.1007/s13201-017-0629-y>.
- Mouna, K., Moncef, G., Rachida, B., 2012. Use of geographical information system and water quality to assess groundwater quality in El Khairat deep aquifer (Enfidha, central East Tunisia). *Arab. J. Geosci.* 5, 1379–1390. <https://doi.org/10.1007/s12517-011-0292-9>.
- Muralidhara Reddy, B., Sunitha, V., Prasad, M., Sudharshan Reddy, Y., Ramakrishna Reddy, M., 2019. Evaluation of groundwater suitability for domestic and agricultural utility in semi-arid region of Anantapur, Andhra Pradesh state, South India. *Groundw. Sustain. Dev.* 9, 100262. <https://doi.org/10.1016/j.gsd.2019.100262>.
- Nagaraja Rao, B.K., Rajurkar, S.T., Ramalingaswamy, G., Ravindra Babu, B., 1987. *Stratigraphy, structure and evolution of the Cuddapah Basin*. *Geol. Soc. India*. 6, 33–86.
- O'Donnell, T.A., Bailer, J.C., Emeleus, H.J., Nyholm, R., 1973. *Fluorine. Comprehensive Inorganic Chemistry Oxford*. 2. Pergamon Press, UK, pp. 1009–1106.
- Panagiotis, P., Ioannis, K., Eleni, V., 2019. Hydrogeochemical assessment and suitability of groundwater in a typical Mediterranean coastal area: a case study of the Marathon basin, NE Attica, Greece. *HydroResearch*. 2, 49–59. <https://doi.org/10.1016/j.hydres.2019.11.002>.
- Pradhan, S.K., Patnaik, D., Rout, S.P., 2001. Water quality index for the groundwater in and around a Phosphatic fertilizer plant. *Indian J. Envi. Protect.* 21, 355–358.
- Prasad, M., Sunitha, V., Sudharshan Reddy, Y., Suvarna, B., Muralidhar Reddy, B., Ramakrishna Reddy, M., 2019. Data on Water Quality Index Development for Groundwater Quality Assessment from Obulavaripalli Mandal, YSR District, A.P India. Data in brief. <https://doi.org/10.1016/j.dib.2019.103846>.
- Qian, H., Li, P., 2011. Hydrochemical characteristics of groundwater in Yinchuan plain and their control factor. *Asian J. Chem.* 23 (7), 2927–2938.
- Qian, H., Li, P., Howard, K.W.F., 2012. Assessment of groundwater vulnerability in the Yinchuan plain, Northwest China using OREADIC. *Environ. Monit. Assess.* 184 (6), 3613–3628. <https://doi.org/10.1007/s10661-011-2211-7>.
- Raghunath, H.M., 1987. *Groundwater*. 2nd edn. New Delhi, Wiley Eastern Limited, p. 563.
- Raju, T., Srimanta, G., Harjeet, K., Raju, B., 2019. Assessment of groundwater quality scenario in respect of fluoride and nitrate contamination in and around Gharbar village, Jharkhand, India. *HydroResearch*. 2, 60–68. <https://doi.org/10.1016/j.hydres.2019.09.002>.
- Rao, N.S., 2009. Fluoride in groundwater, Varaha river basin, Visakhapatnam District, Andhra Pradesh, India. *Environ. Monit. Assess.* 152, 47–60.
- Ravikumar, P., Somashekar, R.K., Augami, M., 2011. Hydrogeochemistry and evaluation of groundwater suitability for irrigation and drinking purposes in the Markendeya river basin, Belugam District, Karnataka state, India. *Environ. Monit. Assess.* 173, 459–487.
- Rittenhouse, G.T., 1961. *The Earths Problem Climates*. University of wisconsin press, Madison, Wisconsin.
- Satyajit, K.G., Ajaykumar, K.K., Ritish, R.R., Akanksha, S.K., Vasant, M.W., Avinash, M.K., Suryakant, P.G., Ramdas, B.M., Namdev, J.P., Kishor, D., KambleSatyajit, K.G., Ajaykumar, K.K., Ritish, R.R., Akanksha, S.K., Vasant, M.W., 2020. Assessment of the groundwater geochemistry from a part of west coast of India using statistical methods and water quality index. *HydroResearch* 3, 48–60. <https://doi.org/10.1016/j.hydres.2020.04.001>.
- Sawyer, C.N., McCarty, P.L., Parkin, G.F., 2003. *Chemistry for Environmental Engineering and Science*. 5th edn. McGraw-Hill, New York, p. 752.
- Singh, H., 1983. *Crop production in India. Agric. Situate India*. 38, 635–639.
- Sreedhar, Y., Nagaraju, A., 2017. Groundwater quality around Tummalapalle area, Cuddapah District, Andhra Pradesh, India. *Appl Water Sci* 7, 4077–4089. <https://doi.org/10.1007/s13201-017-0564-y>.
- Subba Rao, N., 1997. Studies on water quality index in hard terrain of Guntur district, A.P, India. *National Seminar on Hydrology of Precambrian Terrains and Hard Rock Areas*, pp. 129–134.
- Subba Rao, N., Prakasa Rao, J., John Devadas, D., Srinivasa Rao, K.V., Krishna, C., Nagamalleswara Rao, B., 2002. Hydrogeochemistry and groundwater quality in a developing urban environment of a semi-arid region, Guntur, Andhra Pradesh. *J. Geol. Soc. India* 59, 159–166.
- Subramani, T., Elango, L., Damodarasamy, S.R., 2005. Groundwater quality and its suitability for drinking and agricultural use in Chithar River basin, Tamil Nadu, India. *Environ. Geol.* 47, 1099–1110. <https://doi.org/10.1007/s00254-005-1243-0>.
- Sudharshan Reddy, Y., Sunitha, V., Vangala, S., Badri, 2020. Monitoring of groundwater quality for drinking purposes using the WQI method and its health implications around inactive mines in Vemula-Vempalli region, Kadapa District, South India. *Appl Water Sci* 10, 202. <https://doi.org/10.1007/s13201-020-01284-2>.
- Sunitha, V., Sudharshan Reddy, Y., 2019. Hydrogeochemical evaluation of groundwater in and around Lakkireddipalli and Ramapuram, Y.S.R District, Andhra Pradesh, India. *HydroResearch* <https://doi.org/10.1016/j.hydres.2019.11.008>.
- Suvarna, B., Sudharshan Reddy, Y., Sunitha, V., Muralidhar Reddy, B., Prasad, M., Ramakrishna Reddy, M., 2018. Data on application of water quality index method for appraisal of water quality in around cement industrial corridor, Yerraguntla Mandal, Y.S.R District, A.P South India. Data in Brief 28, 104872. <https://doi.org/10.1016/j.dib.2019.104872>.
- Umer, A., Assefa, B., Fito, J., 2019. Spatial and seasonal variation of lake water quality: Beseka in the rift valley of Oromia region, Ethiopia. *Int. J. Energy Water Res.* <https://doi.org/10.1007/s42108-019-00050-8>.
- US Geological Survey, 2000. *Classification of Natural Ponds and Lakes*. US Department of the Interior. US Geological Survey, Washington, DC.
- USEPA, 1993. *Well Land Protection: A Guide for Small Communities Office of Research and Development Office of Water*, Washington, DC (EPA/625/R-93/002).
- WHO, 1983. *Guidelines for Drinking Water Quality*. 3rd ed. World Health Organization, Geneva 2004.
- WHO, 1984. *Guidelines for drinking water quality. Health Criteria and other Supporting Information*. vol 12. World Health Organization, Geneva.
- WHO, 2011. *WHO Guidelines for Drinking-water Quality*. fourth ed. World Health Organization.
- World Health Organization, 2006. *Guidelines for drinking water quality. Recommendations 3rd ed*. Vol 1. Geneva.
- Wu, J., Li, P., Qian, H., 2014. Assessment of soil salinization based on slow cast method and its influencing factors in a semi arid agriculture area, Northwest China. *Environ. Earth Sci.* 71 (8), 3465–3475. <https://doi.org/10.1007/s12665-013-2736-x>.

- Wu, J., Wang, L., Wang, S., 2017. Spatiotemporal variation of groundwater quality in an arid area experiencing long term paper waste water irrigation, Northwest China. *Environ. Earth Sci.* 76 (13), 460. <https://doi.org/10.1007/s12665-017-6787-2>.
- Yidana, S.M., Yidana, A., 2010. Assessing water quality using water quality index and multivariate analysis. *Environ. Earth Sci.* 59, 1461–1473.
- Zhang, Y., Wu, J., Xu, B., 2018. Human health risk assessment of groundwater nitrogen pollution in Jinghui canal irrigation area of the loess region, Northwest China. *Environ. Earth Sci.* 77 (7), 273. <https://doi.org/10.1007/s12665-018-7456-9>.